



JKBOSE 2026

CLASS 11TH

LINEAR INEQUALITIES

MATHS

LECTURE- 04

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TOPICS TO BE COVERED

1

Miscellaneous Exercise

A large, 3D rectangular frame with a red face and yellow borders, tilted slightly to the right. It has a dashed pink border on its face.

**MISCELLANEOUS
EXERCISE ON
CHAPTER 5**

QUESTION

Solve the inequalities in Exercises 1 to 6.

$$2 \leq 3x - 4 \leq 5$$

(Sol) Given that $2 \leq 3x - 4 \leq 5$.

Case 1st: $2 \leq 3x - 4$

$$\Rightarrow 2 + 4 \leq 3x$$

$$\Rightarrow 3x \geq 6$$

$$\Rightarrow x \geq 2$$

$$\Rightarrow x \in [2, \infty) \text{ --- (1)}$$

Case 2nd: $3x - 4 \leq 5$

$$\Rightarrow 3x \leq 5 + 4$$

$$\Rightarrow 3x \leq 9$$

$$\Rightarrow x \leq 3$$

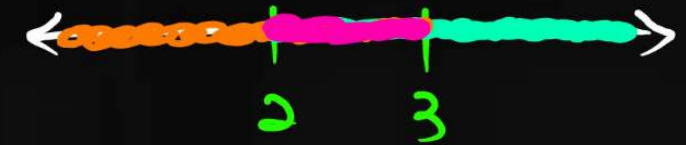
$$\Rightarrow x \in (-\infty, 3] \text{ --- (2)}$$

Therefore from (1) & (2), we get;

$$x \in (-\infty, 3] \cap [2, \infty)$$

$$\Rightarrow x \in [2, 3]$$

So, Solution Set = $[2, 3]$.



QUESTION

Solve the inequalities in Exercises 1 to 6.

$$6 \leq -3(2x - 4) < 12$$

HP/W

QUESTION

Solve the inequalities in Exercises 1 to 6.

$$-3 \leq 4 - \frac{7x}{2} \leq 18$$

h/w.

QUESTION

Solve the inequalities in Exercises 1 to 6.

$$-15 < \frac{3(x-2)}{5} \leq 0$$

H/W.

QUESTION

Solve the inequalities in Exercises 1 to 6.

$$7 \leq \frac{(3x + 11)}{2} \leq 11$$

H/W

QUESTION

Solve the inequalities in Exercises 7 to 10 and represent the solution graphically on number line.

$$5x + 1 > -24, \quad 5x - 1 < 24$$

H/W.

QUESTION



Solve the inequalities in Exercises 7 to 10 and represent the solution graphically on number line.

$$2(x - 1) < x + 5$$

$$3(x + 2) > 2 - x$$

(Sol) Consider the following:

$$2(x - 1) < x + 5$$

$$\Rightarrow 2x - 2 < x + 5$$

$$\Rightarrow 2x - x < 5 + 2$$

$$\Rightarrow x < 7$$

$$\Rightarrow x \in (-\infty, 7) \text{ --- (1)}$$

Also, Consider $3(x + 2) > 2 - x$.

$$\Rightarrow 3x + 6 > 2 - x$$

$$\Rightarrow 3x + x > 2 - 6$$

$$\Rightarrow 4x > -4$$

$$\Rightarrow x > -\frac{4}{4} = -1$$

$$\Rightarrow x > -1$$

$$\Rightarrow x \in (-1, \infty) \text{ --- (2)}$$



From (1) & (2), we get;

$$x \in (-\infty, 7) \cap (-1, \infty)$$

$$\Rightarrow x \in (-1, 7)$$

Therefore the solution set = $(-1, 7)$

Hence the result.

QUESTION

Solve the inequalities in Exercises 7 to 10 and represent the solution graphically on number line.

$$3x - 7 > 2(x - 6), \quad 6 - x > 11 - 2x$$

HP/W

QUESTION

Solve the inequalities in Exercises 7 to 10 and represent the solution graphically on number line.

$$5(2x - 7) - 3(2x + 3) \leq 0, \quad 2x + 19 \leq 6x + 47.$$

H/w.

QUESTION

A solution is to be kept between 68°F and 77°F . What is the range in temperature in degree Celsius (C) if the Celsius/Fahrenheit (F) conversion formula is given by

$$F = \frac{9}{5}C + 32$$

(Sol) Given that $68^{\circ} < F < 77^{\circ}$

$$\Rightarrow 68^{\circ} < \frac{9C}{5} + 32 < 77^{\circ}$$

$$\Rightarrow 68^{\circ} - 32^{\circ} < \frac{9C}{5} + \cancel{32^{\circ}} - \cancel{32^{\circ}} < 77^{\circ} - 32^{\circ}$$

$$\Rightarrow 36 < \frac{9C}{5} < 45$$

$$\Rightarrow \frac{\overset{4}{\cancel{36}} \times 5}{\underset{9}{\cancel{9}}} < C < \frac{\overset{5}{\cancel{45}} \times 5}{\underset{9}{\cancel{9}}}$$

$$\Rightarrow 20^{\circ} < C < 25^{\circ}$$

Therefore the required range is $20^{\circ}\text{C} - 25^{\circ}\text{C}$.

Hence the result.

QUESTION

A solution of 8% boric acid is to be diluted by adding a 2% boric acid solution to it. The resulting mixture is to be more than 4% but less than 6% boric acid. If we have 640 litres of the 8% solution, how many litres of the 2% solution will have to be added?

(Sol) let the number of litres of 2% Boric acid solution to be added = x litres.
So, the resulting mixture = $(640 + x)$ litres.

Then, according to the question;

$$(8\% \text{ of } 640) + (2\% \text{ of } x) > 4\% \text{ of } (640 + x)$$

$$\Rightarrow \frac{8}{100} \times 640 + \frac{2x}{100} > \frac{4}{100} \times (640 + x)$$

$$\Rightarrow \cancel{100} \left(\frac{8}{\cancel{100}} \times 640 + \frac{2x}{\cancel{100}} \right) > \cancel{100} \times \frac{4}{\cancel{100}} (640 + x)$$

$$\Rightarrow 8 \times 640 + 2x > 4 \times 640 + 4x$$

$$\Rightarrow 2x - 4x > 4 \times 640 - 8 \times 640$$

$$\Rightarrow -2x > -4 \times 640$$

$$\Rightarrow x < \frac{4 \times 640}{2} = 1280$$

$$\Rightarrow \boxed{x < 1280} \text{ --- (1)}$$

Also, $(8\% \text{ of } 640) + (2\% \text{ of } x) < 6\% \text{ of } (640+x)$

$$\Rightarrow \frac{8}{100} \times 640 + \frac{2}{100} \times x < \frac{6}{100} (640+x)$$

$$\Rightarrow \cancel{100} \left(\frac{8}{\cancel{100}} \times 640 + \frac{2}{\cancel{100}} x \right) < \cancel{100} \times \frac{6}{\cancel{100}} (640+x)$$

$$\Rightarrow 8 \times 640 + 2x < 6 \times 640 + 6x$$

$$\Rightarrow 2x - 6x < 6 \times 640 - 8 \times 640$$

$$\Rightarrow -4x < -2 \times 640$$

$$\Rightarrow x > \frac{-2 \times 640}{-4} = 320$$

$$\Rightarrow \boxed{x > 320} \text{ --- (2)}$$

From (1) & (2), we get

$$\boxed{320 < x < 1280}$$

So, the required number of litres is greater than 320L & less than 1280L.

Hence the result.

QUESTION

How many litres of water will have to be added to 1125 litres of the 45% solution of acid so that the resulting mixture will contain more than 25% but less than 30% acid content?

(Sol) let the number of litres of water to be added = x litres.

So, the resulting mixture = $(1125+x)$ litres.

Now, according to the question;

$$(45\% \text{ of } 1125) + (0\% \text{ of } x) > 25\% \text{ of } (1125+x)$$

$$\Rightarrow \frac{45}{100} \times 1125 > \frac{25}{100} (1125+x)$$

$$\Rightarrow 45 \times 1125 > 25 \times 1125 + 25x$$

$$\Rightarrow 45 \times 1125 - 25 \times 1125 > 25x$$

$$\Rightarrow 25x < 20 \times 1125$$

$$\Rightarrow x < \frac{20 \times 1125}{25} = 900$$

$$\Rightarrow x < 900 \quad \text{--- 1}$$

Also, $(45\% \text{ of } 1125) + (0\% \text{ of } x) < 30\% \text{ of } (1125+x)$

$$\Rightarrow \frac{45}{100} \times 1125 < \frac{30}{100} (1125+x)$$

$$\Rightarrow 45 \times 1125 < 30 \times 1125 + 30x$$

$$\Rightarrow 45 \times 1125 - 30 \times 1125 < 30x$$

$$\Rightarrow 15 \times 1125 < 30x$$

$$\Rightarrow 30x > 1125 \times 15.$$

$$\Rightarrow x > \frac{1125 \times 15}{30} = 562.5.$$

$$\Rightarrow x > 562.5 \quad \text{--- (2)}$$

From (1) & (2), we get

$$562.5 < x < 900$$

Therefore the no. of litres of water to be added is greater than 562.5 L & less than 900 L.

Hence the result.

QUESTION

IQ of a person is given by the formula

$$CA = 12$$

$$IQ = \frac{MA}{CA} \times 100,$$

where MA is mental age and CA is chronological age. If $80 \leq IQ \leq 140$ for a group of 12 years old children, find the range of their mental age.

$$80 \leq IQ \leq 140$$

$$\Rightarrow 80 \leq \frac{MA}{12} \times 100 \leq 140$$

$$\Rightarrow \frac{80 \times 12}{100} \leq MA \leq \frac{140 \times 12}{100}$$

EXAMPLE

$$|x|^2 = (\pm x)^2 = x^2.$$

$$|x| = \pm x.$$

Solve the inequality: $x^2 - 4|x| + 3 < 0$

(Sol) Given that $x^2 - 4|x| + 3 < 0$

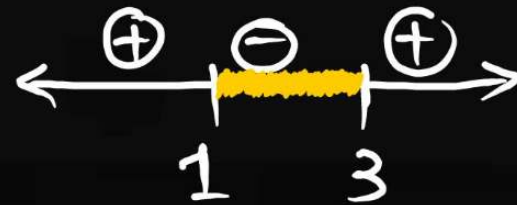
$$\Rightarrow |x|^2 - 4|x| + 3 < 0$$

Here put $|x| = t$, we get;

$$t^2 - 4t + 3 < 0$$

$$\Rightarrow (t-3)(t-1) < 0$$

So, by wavy-curve method, we get
 $t \in (1, 3)$



$$\Rightarrow |x| \in (1, 3)$$

$$\Rightarrow x \in (1, 3) \cup (-3, -1)$$

So, the solution set = $(-3, -1) \cup (1, 3)$.

Hence the result.

QUESTION



The number of integral values of x satisfying the inequality $x^2 - 5|x| + 4 \leq 0$ is

1 $\times$ 4

2 $\times$ 6

3 $\times$ 10

✓ 4 8

(Sol) Given that $x^2 - 5|x| + 4 \leq 0$

$$\Rightarrow |x|^2 - 5|x| + 4 \leq 0$$

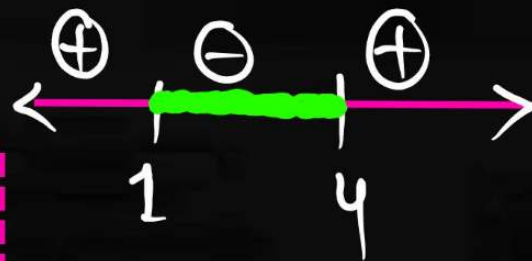
Put $|x| = t$, we get

$$t^2 - 5t + 4 \leq 0$$

$$\Rightarrow (t-1)(t-4) \leq 0$$

$$\Rightarrow t \in [1, 4]$$

$$\Rightarrow |x| \in [1, 4]$$



$$\Rightarrow x \in [1, 4] \cup [-4, -1]$$

Therefore the solution set = $[1, 4] \cup [-4, -1]$

So, the required integers are $-4, -3, -2, -1, 1, 2, 3, 4$

$\Rightarrow$ required no. of integers = 8.

EXAMPLE

$$[-10] = -10; [2] = 2, [2.1] = 2, [10.2] = 10. \\ [-1.1] = -2.$$



Solve the inequality : $[x]^2 - 4[x] + 3 \leq 0$ where $[x]$ denotes Greatest integer less or equal to x .

(Sol) Given that $[x]^2 - 4[x] + 3 \leq 0$

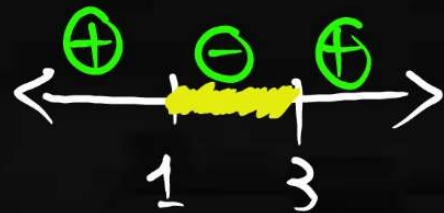
Put $[x] = t$, we get;

$$t^2 - 4t + 3 \leq 0$$

$$\Rightarrow (t-1)(t-3) \leq 0$$

$$\Rightarrow t \in [1, 3]$$

$$\Rightarrow [x] \in [1, 3]$$



$$\Rightarrow [x] = 1, 2, 3.$$

$$\Rightarrow x \in [1, 2), x \in [2, 3), x \in [3, 4)$$

$$\Rightarrow x \in [1, 2) \cup [2, 3) \cup [3, 4)$$

$$\Rightarrow x \in [1, 4)$$

$$\text{So, Solution set} = [1, 4)$$



$$[1] = 1$$

$$[1.2] = 1$$

$$[1.3] = 1$$

$$[2.2] = 2.$$

$$[2.9] = 2$$

$$[2.6] = 2.$$

QUESTION

The number of integral values of x satisfying the inequality $[x]^2 - 9[x] + 20 \leq 0$ is

(Sol) Given that $[x]^2 - 9[x] + 20 \leq 0$

$$\Rightarrow t^2 - 9t + 20 \leq 0, \quad t = [x]$$

$$\Rightarrow (t-4)(t-5) \leq 0$$

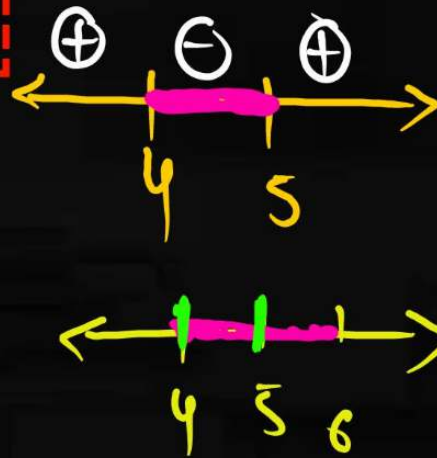
$$\Rightarrow t \in [4, 5]$$

$$\Rightarrow [x] \in [4, 5]$$

$$\Rightarrow [x] = 4, 5$$

$$\Rightarrow x \in [4, 5) \cup [5, 6)$$

$$\Rightarrow x \in [4, 6)$$



So, solution set = $[4, 6)$

So, the req. numbers of integers = 2.

2

3

4

5

Catalan — 6.022×10^{23}

Thank You